Toward On-Board Panoptic Segmentation of Multispectral Satellite Images

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***Abstract*— With tremendous advancements in low-power embedded computing devices and remote sensing instruments, the traditional satellite image processing pipeline which includes an expensive data transfer step prior to processing data on the ground is being replaced by on-board processing of captured data. This paradigm shift enables critical and time-sensitive intelligence to be acquired in a timely manner on-board the satellite itself. However, at present, the on-board processing of multispectral satellite images is limited to classification and segmentation tasks. Extending this processing to the next logical level, we take the first step toward on-board panoptic segmentation of multispectral satellite images and evaluate the applicability of state-of-the-art panoptic segmentation models to an on-board setting. Panoptic segmentation offers major economic and environmental insights, ranging from yield estimation from agricultural lands to intelligence for complex military applications. Nevertheless, the on-board intelligence extraction poses several challenges due to the loss of temporal observations and the need to generate predictions from a single sample. To address this challenge, we propose a multimodal teacher network with a cross modality attention-based fusion strategy to improve segmentation accuracy by exploiting data from multiple modes. We also propose an online knowledge distillation framework to transfer the knowledge learned by this multimodal teacher network to a unimodal student, which receives only a single frame input, and is more appropriate for an on-board environment. We benchmark our approach against existing state-of-the-art panoptic segmentation models using the PASTIS multispectral panoptic segmentation dataset considering an on-board processing setting. Our evaluations demonstrate a substantial 10.7%, 11.9%, and 10.6% increase in segmentation quality (SQ), recognition quality (RQ), and panoptic quality (PQ) metrics compared to the existing state-of-the-art model when it is evaluated in an on-board processing setting.**

***Index Terms*— Knowledge distillation, multimodality fusion, multispectral image processing, on-board satellite image processing, panoptic segmentation.**

I. Introduction

FOLLOWING the launch of Landsat-1, the first satellite to carry a multispectral remote sensing instrument, multispectral imaging of Earth from space has evolved rapidly. The availability of additional spectral bands via multispectral remote sensing enables better segmentation and differentiation of objects compared to using the visible spectrum alone.

The traditional image analysis process for satellite image processing contains typical stages of: 1) data acquisition; 2) data transfer from satellite to the ground; 3) centralized ground processing; and 4) information extraction [1]. This process is slow, in particular, due to the time required to down-link a high-resolution multiband image, meaning that extracted intelligence may be antiquated for effective decision-making. As such, a new paradigm of algorithms is being introduced [1], [2] which can process captured satellite images on-board. This also avoids the expensive and time-consuming data transmission of the original data, and the limited transmission bandwidth can be used to convey the extracted intelligence. The rapid advancements in low-power embedded graphic processing units (GPUs), such as the NVIDIA Jetson series, have further advanced this line of work [1], [3]. In line with this evolution, we propose to take the on-board processing of satellite-acquired multispectral images to the next logical level. Expanding from classification [4], [5] and segmentation tasks [6], [7], we seek to perform panoptic segmentation using limited on-board resources.

Since its first introduction by Kirillov et al. [8], panoptic segmentation has gained substantial traction in numerous computer vision applications. Specifically, panoptic segmentation has helped to eliminate the dichotomy between the recognition of countable objects such as people and animals, and the segregation of uncountable *stuff*, such as sky and roads, by proposing a unified framework to achieve both. Panoptic segmentation is essentially a multitask architecture which jointly performs semantic segmentation and object detection (or instance segmentation), where the former is used for recognizing uncountable, while the latter is employed to segregate countable objects.

Panoptic segmentation enables unmet analytic capabilities in remote sensing, ranging from monitoring urban areas and agricultural lands to military applications [9]. For instance, while semantic segmentation outputs can be used to obtain a general overview of the yield from individual crop types, instance segmentation of agricultural parcels offers an in depth analysis regarding the yield from individual parcels. Similarly, in urban monitoring, advanced intelligence regarding individual households or road/traffic conditions can be obtained by

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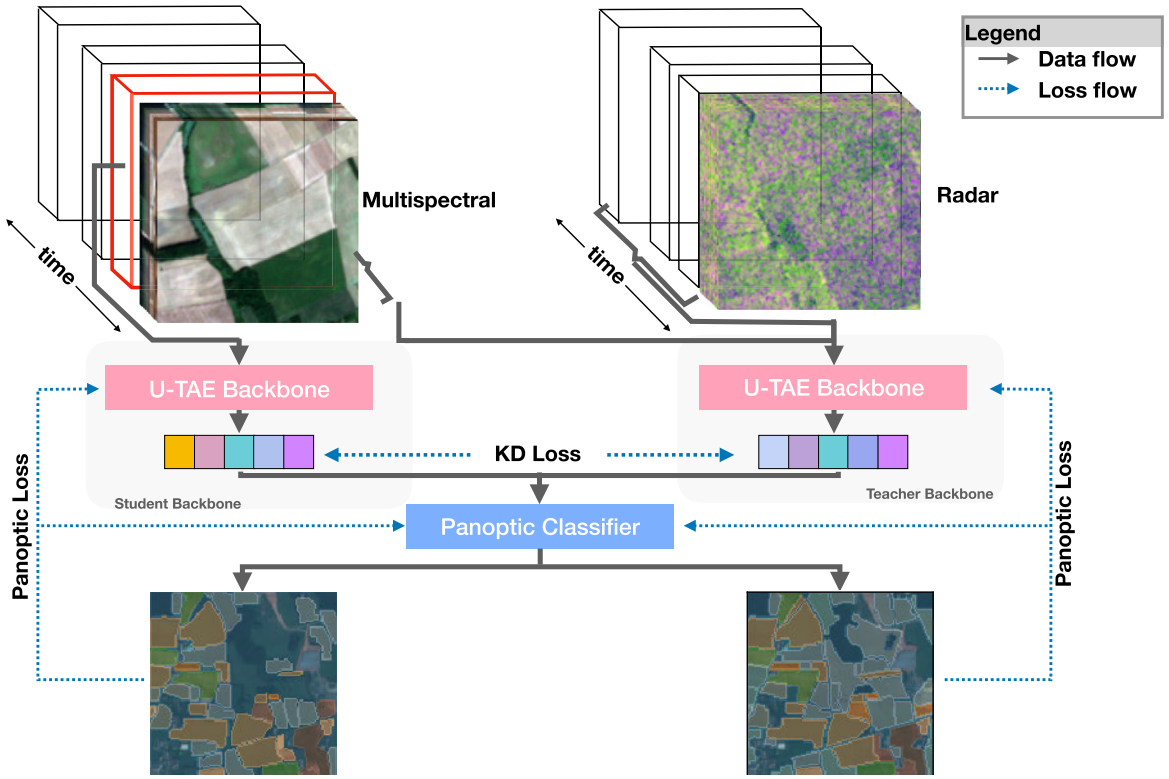


Fig. 1. Proposed cross modality knowledge distillation pipeline. We use a multimodal teacher that receives a time-series input to guide a lightweight, unimodal, single-frame student network.

using panoptic segmentation and considering each detected instance.

In this article, a lightweight framework for multispectral panoptic segmentation is proposed. Specifically, our contributions are threefold. First, we benchmark the performance of several state-of-the-art panoptic segmentation models in an on-board processing setting using the panoptic agricultural satellite timeseries (PASTIS) [10] dataset. To the best of our knowledge, this is the only publicly available multispectral dataset with panoptic annotation. Second, we propose a multimodal fusion architecture to improve panoptic segmentation performance in the on-board computation setting. Third, we propose a novel cross modality knowledge distillation pipeline to transfer the knowledge from this aforementioned multimodal teacher network to a lightweight unimodal student, such that it achieves superior performance. This framework is visually illustrated in Fig. 1. We acknowledge that we are not proposing a complete solution for on-board panoptic segmentation, but rather taking the first step toward achieving such a framework by removing the temporal (i.e., requires time-series data) and modality (i.e., requires multiple modalities that are captured from different satellites) based architectural constraints that existing state-of-the-art methods in the panoptic segmentation domain impose. Furthermore, we have conducted an experiment using Nvidia Jetson Nano GPU to demonstrate the viability of the proposed framework in a resource-constrained setting.

The rest of the article is organized as follows. In Section II we summarize recent developments in related areas and illustrate how the proposed work deviates from these. Section III describes the details of the datasets used, the architectures of our multimodal and unimodal networks, and the knowledge distillation pipeline. Section IV presents the experimental results and conclusions are drawn in Section V.

II. Related Works

A. On-Board Processing of Satellite Images

Since the launch of the Quickbird satellite with high-resolution cameras and high capacity on-board processing capabilities, there has been an increase in research and development related to advanced on-board satellite image processing. Numerous works have investigated on-board capabilities of object detection [1], [11], image classification [12], image selection [13], denoising and artifact removal [14], and change detection [15], [16].

Specifically, considering the most relevant literature in the domains of on-board object detection and classification, Arechiga et al. [11] benchmark a lightweight CNN-based classifier for ship detection using an Nvidia Jetson TX2 GPU. The authors demonstrated the utility of such a CNN architecture over traditional machine-learning techniques. More recently, A you only look once v4 (YOLOv4)-tiny [17] architecture has been benchmarked by [1] for ship detection using HISEA-1 SAR images. The constant false alarm rate (CFAR) algorithm is used to first extract target regions for deep learning-based ship detection. This significantly reduces the number of image patches submitted to the computationally complex deep

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learning-based ship detection algorithm. The detected targets from the CFAR algorithm are stored as 256 × 256 pixel patches which are fed to the YOLOv4-Tiny object detector. Finally, the outputs of the detector are mapped to the original SAR image coordinates using bilinear interpolation. As such, only relevant intelligence is transferred via the satellite-earth data transmission system. In a similar line of work [12] applied transfer learning to adapt a MobileNetV2 [18] classifier pretrained on ImageNet [19] for the classification of earth-observation imagery at different ground-resolved distances. In addition, we would like to acknowledge the Phi-Sat 1 mission, which is the first mission to demonstrate on-board AI capabilities. As illustrated in [20], Phi-Sat 1 hyperspectral images obtained from the HyperScout-2 sensor are used to evaluate cloud segmentation on an Intel Movidius Myriad 2 VPU. Specifically, a convolution encoder-decoder-based architecture named CloudScout is trained to detect and segment cloud regions in the hyperspectral images. To mitigate the hardware limitations of the Myriad 2 VPU, the authors quantize the weights to 16-bit floating points, the input size is reduced to 192 × 192 × 3 and the output segmentation mask is limited to 96 × 96.

We would also such as to note the seminal work conducted by Ziaja et al. [21], which proposes an approach for benchmarking state-of-the-art models for a range of space applications including, crater detection, image classification, and hyperspectral image segmentation. Benchmarking was conducted using a Leopard DPU Evalboard to mimic the on-board environment. Deep Earth (for hyperspectral image segmentation), Deep Mars (for Martian object classification), and Deep Moon (crater detection) models have been validated.

However, in comparison to object detection, semantic segmentation tasks explored to date in on-board processing scenarios are significantly simpler than the panoptic segmentation task we attempt. To the best of our knowledge, no existing work has investigated the use of panoptic segmentation models, which are very computationally complex, in a resource-constrained setting.

A complementary line of research [22], [23], [24], [25] has focused on hardware and software acceleration techniques to enable efficient on-board processing of satellite images. However, these methods optimize matrix multiplication operations and standard convolution and other neural network architecture computations, enabling operation over compressed inputs, and do not propose a specific architecture for the on-board computation. Therefore, we do not review these works in detail, however, acknowledge the contributions of these works toward enabling efficient on-board AI capabilities.

B. Knowledge Distillation

Cross modality knowledge distillation has recently gained attention, but there are limited works concerning the transfer of knowledge from a multimodal teacher to a unimodal student [26]. Specifically, Garcia et al. [27] has used depth and RGB images to train a multimodal teacher while Gao et al. [28] has used multimodal medical images as the input to the teacher. Furthermore, to the best of our knowledge, none of the existing works in cross-modality distillation have used on-line distillation.

In on-line distillation, both teacher and student networks are simultaneously updated, creating an end-to-end trainable pipeline. However, such mutual learning (or joint learning) in a high capacity environment is an open research question that requires further exploration and careful consideration of the relationships between the student and the teacher [26]. For instance, recently an online knowledge distillation framework that uses an ensemble of students and teacher networks is proposed in [29], and these networks are collaboratively trained. Chen et al. [30] have extended this concept to use auxiliary peers and group leaders such that more supervision is provided to the student and diverse knowledge distillation can occur. However, these application settings are quite different from the proposed on-board multispectral panoptic segmentation setting which has a unique set of challenges and requirements. Using multiple student networks is not feasible due to on-board hardware requirements. As such, by leveraging the multiple modalities that are available for the teacher network we propose to overcome the challenges with our application due to the loss of temporal information.

III. Materials and Methods

A. Datasets

1) PASTIS [10] Dataset: The PASTIS dataset is composed of 2433 multispectral image patches, each with ten spectral bands and of size 128 × 128 pixels, at 10 m/pixel resolution. To the best of our knowledge, this is the only publicly available multispectral satellite image dataset with panoptic annotation. This data has been captured using the Sentinel-2 platform and four Sentinel tiles have been used. In Sentinel-2 each pixel is characterized by 13 spectral bands. The authors of this dataset have used only ten bands and have removed atmospheric bands B01, B09, and B10. To obtain consistent spatial resolution all bands have been spatially re-sampled to a 10 m/pixel resolution with bilinear interpolation. The authors have generated a time-series dataset by stacking all the available observations for each patch between September 2018 and November 2019.

However, the temporal sampling of the data is irregular as the orbital schedule of Sentinel-2 means patches are observed different numbers of times and at different intervals. Furthermore, data providers have not processed the images if there is more than 90% cloud coverage, which leads to irregular length time series. The temporal length of the observations ranges from 33 to 61 samples per sequence.

The dataset authors have used the publicly available French Land Parcel Identification System to retrieve the extent and the content of each parcel when annotating the patches. Within the annotations, each pixel has a semantic label that represents the crop type or the background class. Non-agricultural land is classified as the background class. In total there are 18 crop types, the background class and a void class, resulting in 20 semantic classes. The void class denotes agricultural parcels that contain crops that are not classified. The authors have also proposed a fivefold cross validation scheme that divides

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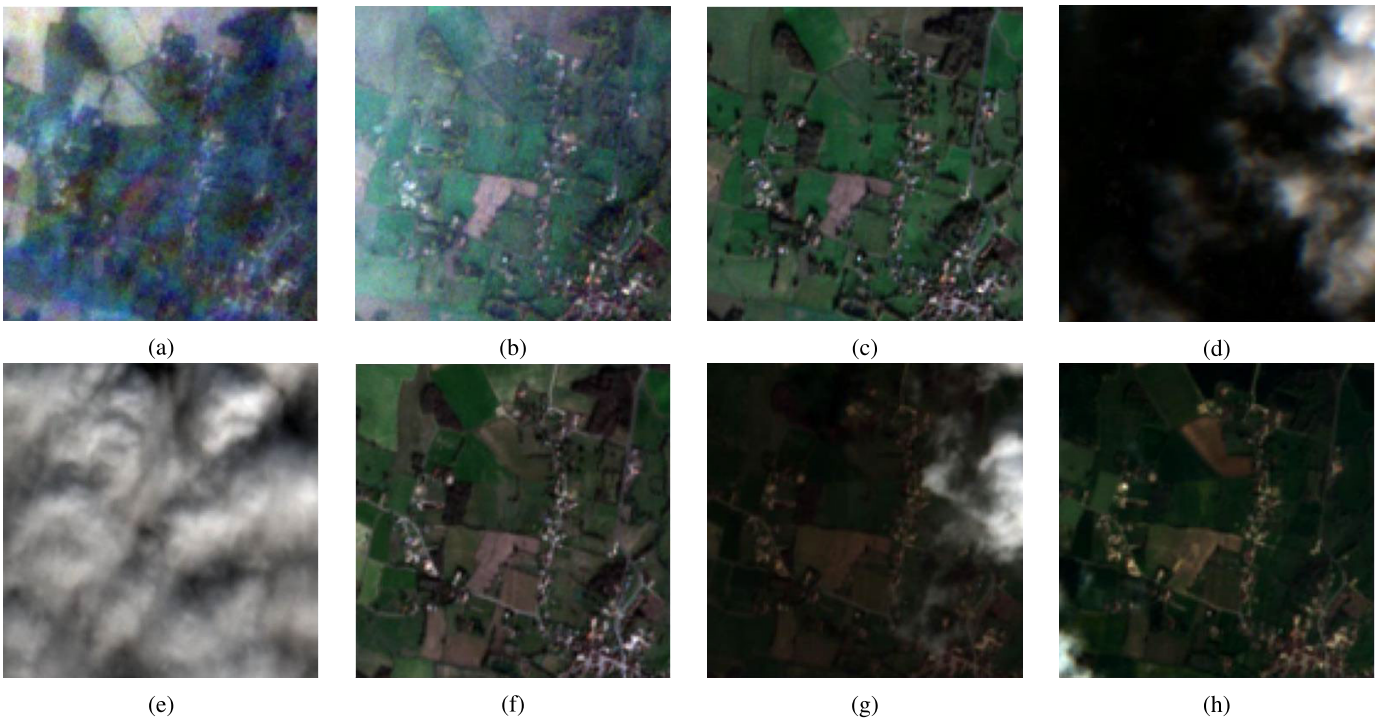


Fig. 2. Visualizations of sample frames from the S2_30634 time series of the PASTIS [10] dataset. The challenges due to cloud cover, illumination conditions and other atmospheric noise is clearly visible in this sequence. (a) Frame 4. (b) Frame 7. (c) Frame 8. (d) Frame 9. (e) Frame 13. (f) Frame 16. (g) Frame 17. (h) Frame 20.

the 2433 patches into five splits. When splitting the data, care was taken to ensure that each fold has patches from all four Sentinel tiles and each split contains comparable class distributions. Furthermore, cross-contamination of data caused by adjacent patches falling into different splits was avoided.

In [10] the authors have observed that even after filtering out predominantly cloudy acquisitions, there still exist partially or completely obstructed patches. Furthermore, they highlight the utility of time-series data for resolving those obstructions and boundary ambiguities due to clouds and other atmospheric noise. For instance, in Fig. 2 we visualize the image sequence of a randomly chosen patch. Within the frames, it can be seen that there exists boundary ambiguities which are indistinguishable even for humans. The multiple temporal observations can be used to eradicate these ambiguities in ground-based processing, however, in an on-board setting an AI algorithm does not have access to such historical data. In particular, when benchmarking the algorithms presented in [10] in a nontemporal (single frame) setting we illustrate how state-of-the-art methods struggle to generate accurate predictions (See Section IV-B). Hence we propose an on-line knowledge distillation strategy to augment a lightweight model that observes only a randomly chosen single frame of the time series, allowing it to *imagine* the corresponding features from the time series and achieve more accurate panoptic segmentation.

2) PASTIS-R [31] Dataset: PASTIS-R is an extension of the PASTIS dataset where the authors have combined the Sentinel-2 multispectral patches with the corresponding Sentinel-1 observations. Specifically, they have used Sentinel-1 data in ground range detected format and assembled these into three-channel images (vertical polarization, horizontal polarization, and the ratio of vertical over horizontal polarization). Separate observations were captured in both ascending and descending orbits, generating two three-channel images for each observation. This multimodal dataset has rich spectral information coming from the Sentinel-2 multispectral observations, while the C-band radar of Sentinel-1 captures useful geometric information as it's immune to cloud cover [31].

Hence, this multimodal dataset can be seen as containing complementary information to be leveraged when views are obstructed due to cloud and other illumination conditions, however, it is illogical to directly fuse the data in an on-board setting as data is captured from separate satellites. Therefore, we propose a knowledge distillation strategy where a heavyweight multi-modal teacher guides a unimodal single frame student such that it can imagine the unseen features which are only visible to the teacher.

B. Knowledge Distillation Framework

In this section, we introduce the structure of the multimodal backbone that we use for our teacher network (Section III-B1) and the architecture of the lightweight unimodal backbone of the student network (Section III-B2). In Section III-B3 we describe how the extracted features from these backbones are used in the panoptic classifier while Section III-B4 outlines the loss functions used to train the network.

1) Teacher Backbone With Cross-Modality Attention: In this section we illustrate how the proposed cross-modality attention backbone is formulated. The approach takes inspiration from the U-TAE backbone from [10], though we incorporate specific augmentations to increase its capabilities in

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a multimodal setting. The spatio-temporal encoding structure is based on [10], however, prior to the decoding phase we employ cross-modality attention to extract task-specific, salient information that compliments each modality.

The inputs to the teacher network, $\dot{X}$ and $\ddot{X}$, are spatio-temporal inputs with shapes $T \times \dot{C} \times W \times H$ and $T \times \ddot{C} \times W \times H$, respectively, where T denotes the length of the time series (i.e., number of frames), W indicates the width of the image and H denotes its height. $\dot{C}$ and $\ddot{C}$ represent the number of channels in the multispectral and radar modalities, respectively. Each spectral encoder, $\dot{f}_e^l$ and $\ddot{f}_e^l$ ($l \in [1, \ldots, L]$), has L levels where each level is comprised of convolutions, Relu and normalization layers. The encoder f_e^l ($f_e^l \in [\dot{f}_e^l, \ddot{f}_e^l]$) at level l receives the output $\dot{e}_t^{l-1}$ ($t \in [1, \ldots, T]$) of the immediately proceeding layer and generates a feature map $\dot{e}_t^l$. At each level, the encoder maps are compressed using strided convolutions such that the output height $H^l = (H/2^{l-1})$, and width $W^l = (W/2^{l-1})$. All frames in the sequence are simultaneously processed by the encoders.

To perform temporal encoding, similar to [10], we use a lightweight temporal attention encoder (L-TAE) [32] which is a multihead self-attention network with G heads. The feature maps $\dot{e}^l$ and $\ddot{e}^l$ of the last level (i.e., L) are used and the L-TAE networks, $\dot{f}_{L-TAE}$ and $\ddot{f}_{L-TAE}$ generate G attention vectors, each of shape $T \times W_L \times H_L$ as follows:

$$\begin{aligned}[\dot{a}^{L,1}, \ldots, \dot{a}^{L,G}] &= \dot{f}_{\text{L-TAE}}(\dot{e}^l) \\ [\ddot{a}^{L,1}, \ldots, \ddot{a}^{L,G}] &= \ddot{f}_{\text{L-TAE}}(\ddot{e}^l).\end{aligned} \tag{1}$$

To apply these attention maps to all encoded maps from the U-Net, interpolation is performed, where each attention vector, $\dot{a}^{l,g}$ and $\ddot{a}^{l,g}$ in g is interpolated to size H_l and W_l for level $l \in [1, \ldots, L]$. The respective channels $\dot{C}$ and $\ddot{C}$ of the multispectral and radar modalities are split into G consecutive groups (i.e., $[\dot{e}^{l,1}, \ldots, \dot{e}^{l,G}]$, each of shape $T \times (\dot{C}_l/G) \times W_l \times H_l$; and $[\ddot{e}^{l,1}, \ldots, \ddot{e}^{l,G}]$, each of shape $T \times (\ddot{C}_l/G) \times W_l \times H_l$).

Then we multiply each embedding by it's respective attention map, average the temporal sequence, and pass the resultant feature vector through a 1×1 convolution layer

$$\begin{aligned}\dot{h}^l &= \dot{f}^l_{\text{Conv}_{1\times1}}\left[\sum_{t=1}^{T} \dot{a}_t^{l,g} \dot{e}_t^{l,g}\right]_{g=1}^{G} \\ \ddot{h}^l &= \ddot{f}^l_{\text{Conv}_{1\times1}}\left[\sum_{t=1}^{T} \ddot{a}_t^{l,g} \ddot{e}_t^{l,g}\right]_{g=1}^{G}\end{aligned} \tag{2}$$

where [.] denotes concatenation along the channel dimension.

In [31], the authors have introduced several fusion strategies for the PASTIS-R dataset. Their evaluations using early, late, decision, and mid level fusion indicated that early fusion is the best strategy for panoptic segmentation. However, leveraging the generated attention vectors, $\dot{a}^{l,g}$ and $\ddot{a}^{l,g}$, we propose fusion of the multispectral and radar modalities using cross-modality attention. Specifically, we generate attention maps, $\dot{a}^l$ and $\ddot{a}^l$ for each modality such that

$$\begin{aligned}\dot{a}^l &= \frac{\left[\sum_{t=1}^{T}\sum_{g=1}^{G} \dot{a}_t^{l,g}\right]}{T \times G} \\ \ddot{a}^l &= \frac{\left[\sum_{t=1}^{T}\sum_{g=1}^{G} \ddot{a}_t^{l,g}\right]}{T \times G}.\end{aligned} \tag{3}$$

These attention maps denote how salient features are spatially arranged in a particular modality. Hence, they are important for capturing complementary information from a second mode. We interpolate these 2-D maps to match the channel dimension of the complementary mode, and generate an augmented feature such that

$$\hat{h}^l = f^l_{\text{Conv}_{1\times1}}\left[\dot{h}^l, \ddot{h}^l, \dot{h}^l\ddot{a}^l, \ddot{h}^l\dot{a}^l\right] \tag{4}$$

where [,] denotes concatenation in the channel dimension. Then the augmented feature $\hat{h}^l$ is used by the spatial decoder.

Following the conventional U-Net network structure, the encoder map from level l, $\hat{h}^l$, is concatenated with the decoded feature map, $\hat{d}^{l-1}$, from level $l-1$. Each decoder block is composed of strided transposed convolutions to upsample the feature maps from the previous level. The decoder output is formally denoted as $\hat{d} = [\hat{d}^1, \ldots, \hat{d}^L]$.

2) Unimodal Single Frame Backbone for the Student: The student backbone receives the input, $\tilde{X}$, of shape $\dot{C} \times W \times H$, that corresponds to a single frame from the multispectral modality. Similar to the teacher backbone we pass it through a spatial encoder $\tilde{f}_e^l$ with L levels and generate the feature map e^l. However, in contrast to the teacher backbone, $\tilde{e}^l$ is of dimension $C_l \times W_l \times H_l$. Despite the absence of the temporal dimension, the L-TAE is used in the student backbone with the motivation of capturing the channel-wise relationships with the aid of a multihead attention formulation such that

$$[\tilde{a}^{L,1}, \ldots, \tilde{a}^{L,G}] = \tilde{f}_{\text{L-TAE}}(\tilde{e}^L) \tag{5}$$

however, when applying the attention no temporal summarization is performed. Specifically

$$\tilde{h}^l = \tilde{f}^l_{\text{Conv}_{1\times1}}\left[\tilde{a}^{l,g}\tilde{e}_t^{l,g}\right]_{g=1}^{G} \tag{6}$$

where [.] denotes concatenation along the channel dimension. Then, similar to the teacher backbone, the spatial decoder of the student up samples these features to generate the multiscale output features, $\tilde{d} = [\tilde{d}^1, \ldots, \tilde{d}^L]$.

3) Panoptic Classifier: The panoptic classifier receives multiscale feature maps (either from the teacher or student backbones) and produces predictions over parcels. We used the same architecture as the panoptic classifier network from [10], where it makes four predictions for each parcel: a center point, a bounding box, a binary instance mask, and a class. We refer the readers to [10] for further details regarding this network's architecture.

We use a single instance of this panoptic classifier to evaluate the predictions of both student and teacher backbones, allowing it to identify the similarities and correspondences between the feature vectors $\hat{d}$ and $\tilde{d}$. Note that in our formulation, the dimensions of the decoded maps from the student and teacher backbones at each level of the U-Net are identical.

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4) Loss Functions for On-Line Knowledge Distillation: As discussed in Section III-B3, the panoptic classifier makes four predictions, a centerness heat map, bounding boxes, instance masks, and classes, and as such [10] used four losses to supervise these predictions.

Formally, let $\breve{m} \in [0,1]^{W\times H}$ denote the ground truth centerness heat map, let m be the predicted centerness heat map, and let there be P ground truth parcels. Then L_{center} is defined as follows:

$$L_{\text{center}} = \frac{-1}{|P|}\sum_{\substack{i=1,\dots,H\\ j=1,\dots,W}} = \begin{cases}\log(m_{i,j}) & \text{if } \breve{m}_{i,j}=1\\ (1-\breve{m}_{i,j})^{\beta}\log(1-m_{i,j}) & \text{else}\end{cases}$$

where β is a hyper-parameter and following [10] it is set to 4.

Then P' is defined as a set of detected parcels and $k'(p)$ denotes the predicted class of the parcel $p \in P'$, while $k''(p)$ indicates its ground truth class. Then the class loss of parcel p is defined as follows:

$$L^{p}_{\text{class}} = -\log(k'(p)[k''(p)]). \tag{7}$$

Let $h'(p)$ and $h''(p)$ denote the predicted and ground truth heights of the bounding boxes of parcel p, and $w'(p)$ and $w''(p)$ indicate its respective width. Then

$$L^{p}_{\text{size}} = \frac{|h'(p)-h''(p)|}{h''(p)} + \frac{|w'(p)-w''(p)|}{w''(p)} \tag{8}$$

defines the size loss. In addition, a pixel-wise binary cross entropy (BCE) loss is computed between the predicted shape, $l'(p)$ and the corresponding ground truth instance $s''(p)$ mask when it is cropped using the predicted bounding box. This can be defined as follows:

$$L^{p}_{\text{shape}} = \text{BCE}(l'(p), \text{crop}(\text{s}''(\text{p}))). \tag{9}$$

Then the complete loss function in [10] can be written as follows:

$$L = L_{\text{center}} + \frac{1}{P'}\sum_{p\in P'}\left(L^{p}_{\text{class}} + L^{p}_{\text{size}} + L^{p}_{\text{shape}}\right). \tag{10}$$

To facilitate an on-line student-teacher knowledge distillation framework we perform the following augmentations. Let f_{panoptic} denote the output of the panoptic classifier introduced in Section III-B3 and let L denote the complete loss defined in (10). Then

$$\begin{aligned} L_{\text{student}} &= L\left(f_{\text{panoptic}}\left(\{\tilde{d}\}_1^L\right)\right)\\ L_{\text{teacher}} &= L\left(f_{\text{panoptic}}\left(\{\hat{d}\}_1^L\right)\right)\\ L_{\text{distill}} &= \sum_{i=1}^{L}||\tilde{d}_i - \left[\hat{d}_i\right]^{\text{detach}}||^2 \end{aligned} \tag{11}$$

where $[\hat{d}_i]^{\text{detach}}$ denotes detaching the tensor from gradient computation. Then the final loss of our framework is defined as follows:

$$L^{*} = L_{\text{student}} + \lambda_1 L_{\text{teacher}} + \lambda_2 L_{\text{distill}} \tag{12}$$

where λ_1 and λ_2 are hyper-parameters controlling the contributions of the respective loss terms.

TABLE I

EVALUATION RESULTS OF THE UNIMODAL AND FUSION MODELS USING THE TIME-SERIES INPUT OF THE PASTIS-R [31] DATASET

Model	SQ	RQ	PQ	Parameters	Run Time
U-TAE+PAPS-Multispec [10]	81.3	49.2	40.4	1.2M	0.53s
U-TAE+PAPS-Radar	77.2	39.1	30.8	1.2M	0.38s
U-TAE+PAPS-Late Fusion [33]	81.6	50.5	41.6	2.3M	0.73s
U-TAE+PAPS-Early Fusion [33]	82.2	50.6	42.0	1.7M	0.52s
Self Attention Fusion	82.3	53.1	43.8	2.4M	0.89s
Cross Attention Fusion	**82.7**	**55.6**	**45.8**	2.4M	0.98s

IV. EVALUATIONS

In Section IV-A we first report the performance of the proposed fusion architecture when the time-series PASTIS-R dataset's input is used. In Section IV-B evaluations of the state-of-the-art panoptic segmentation methods in the on-board setting without time-series data (i.e., single input frame) are provided. In Section IV-C, we evaluate the proposed knowledge distillation pipeline, and in Section IV-F we qualitatively illustrate how the student features have been augmented by the teacher.

Similar to [8] we report class averaged segmentation quality (SQ), recognition quality (RQ), and panoptic quality (PQ), where RQ corresponds to the quality of the detection and can be computed using

$$\text{RQ} = \frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|} \tag{13}$$

where TP, FP, and FN denote true positive, false positive, and false negative, respectively. SQ is computed as the average IoU of matched segments and can be written as follows:

$$\text{SQ} = \frac{\sum_{(i,j)\in \text{TP}}\text{IoU}(i,j)}{|\text{TP}|} \tag{14}$$

where i, j denotes the detected parcels within the set of true positives. Then PQ is the product of SQ and RQ (i.e., PR = SQ $\times$ RQ), thus combining the detection, classification, and delineation accuracies [31].

A. Results of the Fusion Model

In Table I we report the performance of the proposed cross attention-based fusion model together with the trainable parameter counts and run times. Note that the run times are reported for evaluating a single sample with 40 frames in the time series. In addition, we report the performance of the existing baselines and attention variations for comparison. Note that due to sampling rate inconsistencies between the modalities, similar to [31], we have interpolated Radar observations to match the time stamps of the multispectral modality.

Analyzing the results presented in Table I, it is clear that fusion methods have been able to achieve superior performance compared to their unimodal counterparts. Considering the difference between the unimodal multispectral baseline and the proposed Cross Modality Fusion, we observe a more than 5% increase in RQ and PQ metrics. Furthermore, attention-based fusion strategies have achieved a comfortable increase over early and late fusion schemes. Most importantly, the proposed cross attention scheme yields a 0.5%, 5%, and 3.8%

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TABLE II
RESULTS OF THE STATE-OF-THE-ART MODELS USING A SINGLE FRAME INPUT OF THE PASTIS-R [31] DATASET

Model	Modality	SQ	RQ	PQ	Parameters	Runtimes
Detectron 2-FPN R-50 [34]	RGB	40.98	7.43	4.4	46.0M	0.81s
Detectron 2-FPN R-101 [34]	RGB	42.14	5.87	3.6	65.0M	0.85s
U-TAE + PAPS [10]	MutiSpec	63.8	18.8	13.8	1.2M	0.37s
U-TAE + PAPS-Late Fusion [33]	MutiSpec + Radar	73.4	20.5	16.1	2.3M	0.61s
U-TAE + PAPS-Early Fusion [33]	MutiSpec + Radar	74.4	26.4	20.9	1.7M	0.41s
Self Attention Fusion	MutiSpec + Radar	74.9	28.8	23.0	2.4M	0.74s
Cross Attention Fusion	MutiSpec + Radar	**78.1**	**31.6**	**24.7**	2.4M	0.81s

TABLE III
PERFORMANCE OF THE KNOWLEDGE DISTILLATION MODELS USING A SINGLE INPUT FRAME FROM THE PASTIS-R [31] DATASET. STUDENT NETWORKS ONLY RECEIVE THE MULTISPECTRAL MODALITY OF PASTIS-R. SQ, RQ, AND PQ ARE SHOWN FOR THE STUDENT NETWORK

Student	Teacher	SQ	RQ	PQ	Parameters (of student)	Run times (of student)
U-Net + PAPS	U-TAE + PAPS (MultiSpec)	71.4	14.9	11.5	1.1M	0.32s
U-TAE + PAPS	U-TAE + PAPS (MultiSpec)	71.6	17.0	12.7	1.2M	0.37s
U-Net + PAPS	Self-Attention Fusion	70.2	23.0	17.6	1.1M	0.32s
U-TAE + PAPS	Self-Attention Fusion	73.3	27.8	21.9	1.2M	0.37s
U-Net + PAPS	Cross Attention Fusion	73.6	25.7	20.2	1.1M	0.32s
U-TAE + PAPS	Cross Attention Fusion (off-line)	74.5	27.3	21.8	1.2M	0.37s
U-TAE + PAPS	Cross Attention Fusion	**74.5**	**30.7**	**24.4**	1.2M	0.37s

performance gain in SQ, RQ, and PQ metrics, respectively, compared to the previous state-of-the-art early fusion mechanism. This denotes the value of extracting salient spatial features across different modalities for informed decision-making. By understanding which features are available and where they are important, the overall model is better able to adapt to noisy and occluded observations.

B. Benchmarking the State-of-the-Art Models in an On-Board Setting

In Table II we benchmark the baseline models and their attention variants in an on-board setting, where we have used only a single frame from the time-series input. Note that when selecting a frame from the time series, a random frame is selected and in the multimodal setting we have used a pair of frames that correspond to the same timestamp from the two modalities.

Our analysis reveals the challenge of performing panoptic segmentation from a single frame, without the time-series input. Considering the performance degradation between Tables I and II for the U-TAE + PAPS models that receive the multispectral inputs, we observe a significant drop of 17.5%, 30.4% and 26.6% in SQ, RQ, and PQ metrics, respectively. We observe that the use of multiple modes helps mitigate the performance drop, however, we also highlight the lack of practicality in adapting such a framework to an on-board evaluation setting given the limited on-board computing power. Therefore, utilizing the cross modality attention fusion model that receives a time-series input as our teacher network, we seek to augment the low capacity U-TAE + PAPS network (that receives a single frame from the multispectral time-series input).

C. Knowledge Distillation Results

In Table III, we report the performance of different configurations of student and teacher networks. Note that when choosing the configurations for the student network, we do not consider detectron2, nor do we consider late and early fusion methods for the teacher network, due to the low performance of these networks. Note that for the student network, only the multispectral modality of PASTIS-R was provided, while we have considered teacher networks using both unimodal (multispectral input) and multimodal inputs. In addition, we have used a pretrained teacher network and trained only the student network to evaluate the off-line knowledge distillation setting. The rest of the teacher networks are jointly trained with the student and thus perform on-line knowledge distillation.

Analyzing the results presented in Table III, we observe the value of knowledge distillation across all configurations that we evaluate. Specifically, we see a 10.7%, 11.9%, and 10.6% increase in SQ, RQ, and PQ metrics, respectively, when the multispectral U-TAE + PAPS model is trained with a Cross Attention Fusion teacher, compared to training the U-TAE + PAPS model alone. Furthermore, we observe that on-line knowledge distillation leads to better knowledge transfer compared to the off-line setting, where the teacher model is pretrained and frozen. We believe that this is due to the fact that on-line distillation allows the PAPS classifier to better understand how different networks behave in different evaluation conditions and act accordingly, rather than simply trying to re-create the features of the teacher from the student. Moreover, we note a significant performance gap between the U-Net and the U-TAE backbones, resulting from the introduction of L-TAE. Despite the absence of a temporal time-series input, we observe a better encoding through the L-TAE attention scheme. We would such as to highlight the fact that $f_{\text{L-TAE}}$ [See (1)] is operating on the feature dimension of e^L. As such, even though Garnot and Landrieu [10] have applied L-TAE to generate temporal attention scores for temporal summarization, we can re-purpose it to better scale the features in the backbone network which results in superior performance compared to the traditional U-Net backbone.

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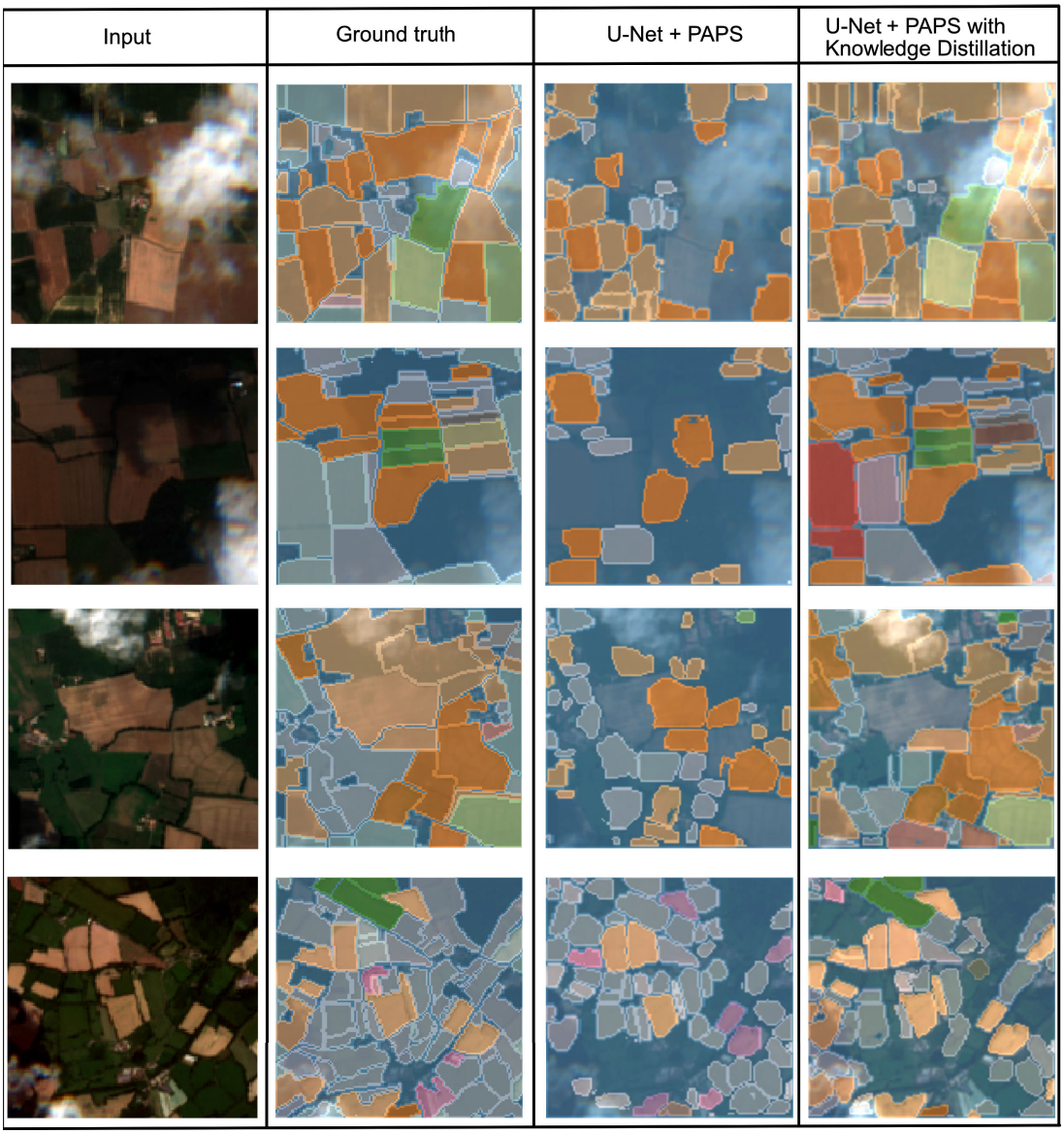


Fig. 3. Qualitative results of the baseline U-TAE + PAPS model of [10] and the U-TAE + PAPS student network trained with the cross att fusion teacher network using the proposed on-line knowledge distillation framework for a single input frame.

To better identify the significance of the performance gap between the naive multispectral U-TAE + PAPS model and the same model trained using the proposed knowledge distillation paradigm we conducted a statistical significance test. We used the fivefold cross-validation splits provided by the dataset authors and trained the model twice across the provided fivefold, yielding ten trials. In each of those trials, the model parameters are randomly initialized and the model is trained from scratch. Using the average value for the PQ metric across the ten trials we conduct a Student's t-test, which resulted in a value of 11.7738 that clearly illustrating that the difference in mean accuracy between the two models is statistically significant.

D. Robustness to Noise

To demonstrate the robustness of the proposed framework to noise, we construct a noisy test set by artificially contaminating the test data using both Gaussian and Impulsive (salt-and-pepper) noise. Gaussian noise is generated with a mean of zero and standard deviation of 0.01, and when generating impulsive noise white and black contamination are selected with equal probability. Similar to [35] we only contaminated the test set and the model has only seen clean data (uncontaminated samples) during the training. From each test sample, we randomly select between 10% and 20% of pixels for noise injection.

Evaluation results on both Gaussian and Impulsive noise injected test sets are given in Table IV. In addition to the PQ metric, we also report the difference between the PQ metric obtained for the uncontaminated test set and the noisy test set in brackets.

The best robustness to noise is achieved with the multimodal architecture (i.e., cross attention fusion), as it receives more information and can better compensate for the noise. When comparing the unimodal models, we observe an improved robustness to noise from U-TAE + PAPS model when it is

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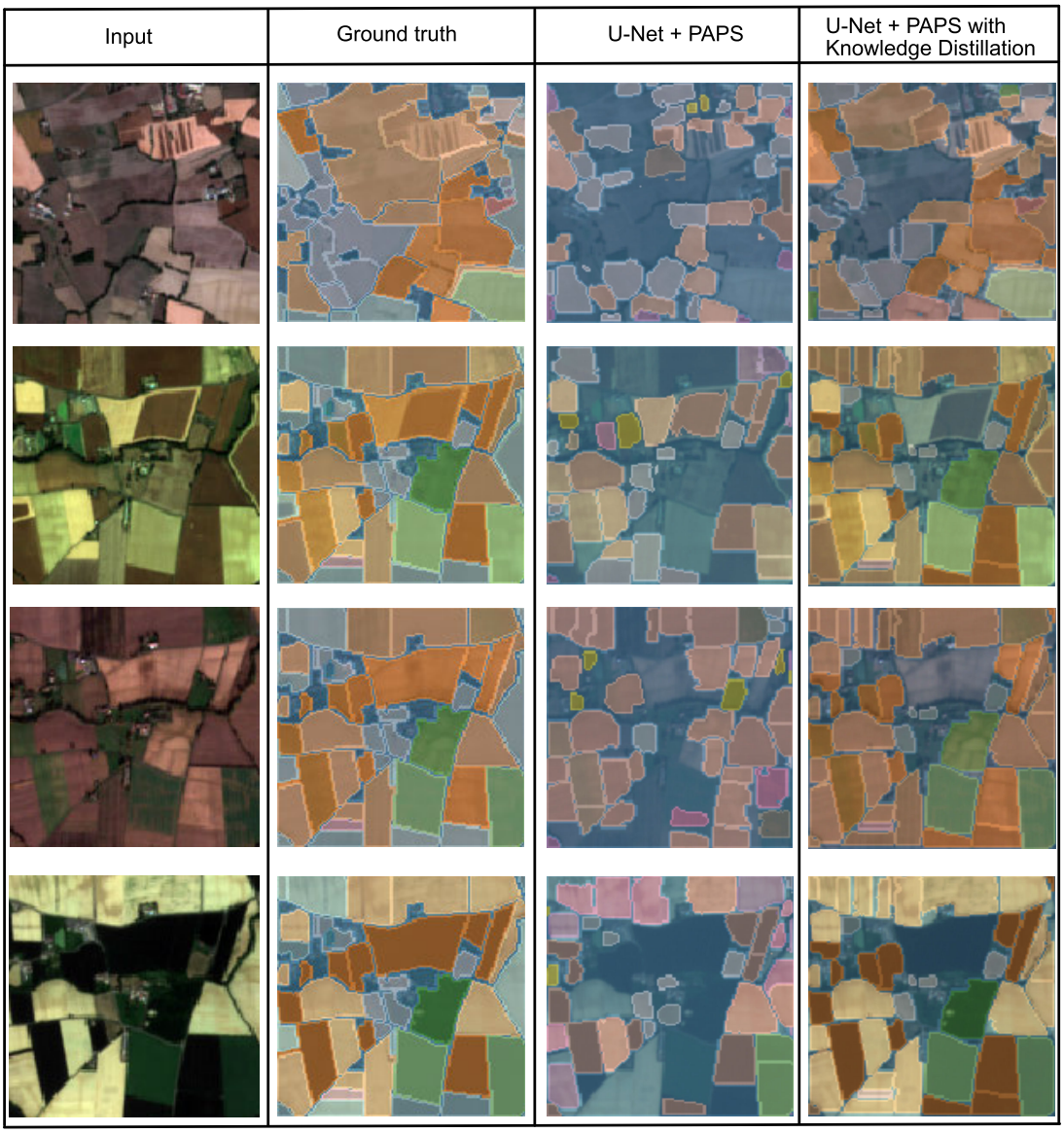


Fig. 4. Additional Qualitative results of the baseline U-TAE + PAPS model of [10] and the U-TAE + PAPS student network trained with the cross att fusion teacher network using the proposed on-line knowledge distillation framework for a single input frame.

TABLE IV

PERFORMANCE USING THE NOISY TEST SETS OF PASTIS-R [31] DATASET CONSTRUCTED USING GAUSSIAN AND IMPULSIVE (SALT-AND-PEPPER) NOISE MODELS. NOTE THAT WE REPORT PQ METRIC TOGETHER WITH THE DIFFERENCE BETWEEN THE PERFORMANCE IN UNCONTAMINATED AND NOISY SETTINGS (WITHIN BRACKETS)

Model	PQ with Gaussian Noise	PQ with Impulsive noise
U-TAE + PAPS	9.5 (4.3)	8.6 (5.2)
U-TAE + PAPS (Student)	21.8 (2.6)	21.4 (3.0)
Cross Attention Fusion	22.1 (2.3)	21.6 (2.8)

trained using the proposed knowledge distillation framework (row 2 compared to row 1), despite models having the same architecture, with the robustness of U-TAE + PAPS (student) is quite close to that of the multimodal architecture. We believe this is a result of the student-teacher setting allowing them to anticipate features from the missing modality, which grant the model increased stability in the presence of noise.

E. Evaluation Using Hardware Constraints

An additional experiment was conducted to evaluate the applicability of the proposed framework to a resource-constrained setting. For this experiment, we used a Nvidia Jetson Nano GPU which is equipped with 4 GB 64-bit LPDDR4 (25 600 MHz) memory and has 5–10 W power consumption. As noted in Section II, the PhiSat-1 and PhiSat-2 missions from the European Space Agency use an Intel Movidius Myriad 2 VPU for onboard processing. The Intel Movidius Myriad VPU has a lower Power consumption of 1–5 W, and possesses 4 GB of LP-DDR3 (733 MHz) Memory. The Movidius VPU uses the OpenVINO toolkit to provide an interface between the deep learning libraries and the VPU, and ONNX is used at an intermediate translation level.

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Fig. 5. Additional Qualitative results of the baseline U-TAE + PAPS model of [10] and the U-TAE + PAPS student network trained with the cross att fusion teacher network using the proposed on-line knowledge distillation framework for a single input frame.

Similar to [21], we encountered problems when translating our models, in particular, due to the custom layers used. As such, we conducted our evaluations using the Nvidia Jetson Nano GPU.

The Jetson Nano is setup with the official operating system provided by Nvidia, Linux4Tegra, is installed on a 32 GB microSD card. Pytorch is installed using a publicly available wheel using the Python package manager (pip). No changes are made for the configuration of our U-TAE + PAPS (student) model, except the batch size which is set to one. Using a batch size of one replicates a real-world test application where the model receives one test image and predictions are generated for that single test image. The model was executed for 100 test iterations (i.e., 100 test images) and the average time to process one image and generate panoptic predictions is reported. On average, our model generates panoptic outputs in 15.90 s.

F. Qualitative Results

In Figs. 3–5 we present qualitative results when using a single frame from the time-series input of PASTIS-R [31]. The results when the multispectral U-TAE + PAPS network is trained with a multimodal, cross modality attention fusion teacher network are presented together with the baseline U-TAE + PAPS model of [10] (trained with a single frame input) for comparisons.

Figs. 3–5 clearly shows that the baseline U-TAE + PAPS network struggles due to cloud cover as well as the illumination inconsistencies and the resultant boundary ambiguities, leading to miss-classifications and missed detection. Correctly identified parcels are significantly fewer compared to the ground truth. Leveraging the time-series input and multimodality observations our cross modality attention fusion teacher network has been able to positively influence the student network, which allowed the same U-TAE + PAPS

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student model to generate comparatively better results through the proposed knowledge distillation pipeline.

V. Conclusion

In this article, we formulated and evaluated different deep neural network architectures for on-board panoptic segmentation of multispectral satellite images. To this end, we first benchmarked existing state-of-the-art panoptic segmentation methodologies in an on-board setting and observed the performance degradation that occurs, largely due to loss of temporal data, which is needed to reduce confusion and ambiguities that arise due to cloud obstruction and other atmospheric noise. We seek to compensate for this loss of information through the introduction of an additional modality, and an augmented multimodality fusion strategy is proposed using a cross-modality attention fusion scheme. Most importantly, we alleviated additional information processing burden caused by the introduction of multimodality streams via a knowledge distillation pipeline, where a heavy-weight multimodality teacher that receives a time-series input was used to guide a lightweight unimodal student network which only receives a single frame from the multispectral time series. With this approach, a significant 10.7%, 11.9%, and 10.6% increase in SQ, RQ, and PQ metrics were observed compared to training the baseline U-TAE + PAPS single-frame unimodal model alone. In future work, we will investigate methods to further augment the efficiency and the performance of our student model while extending the framework to process hyperspectral images.

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